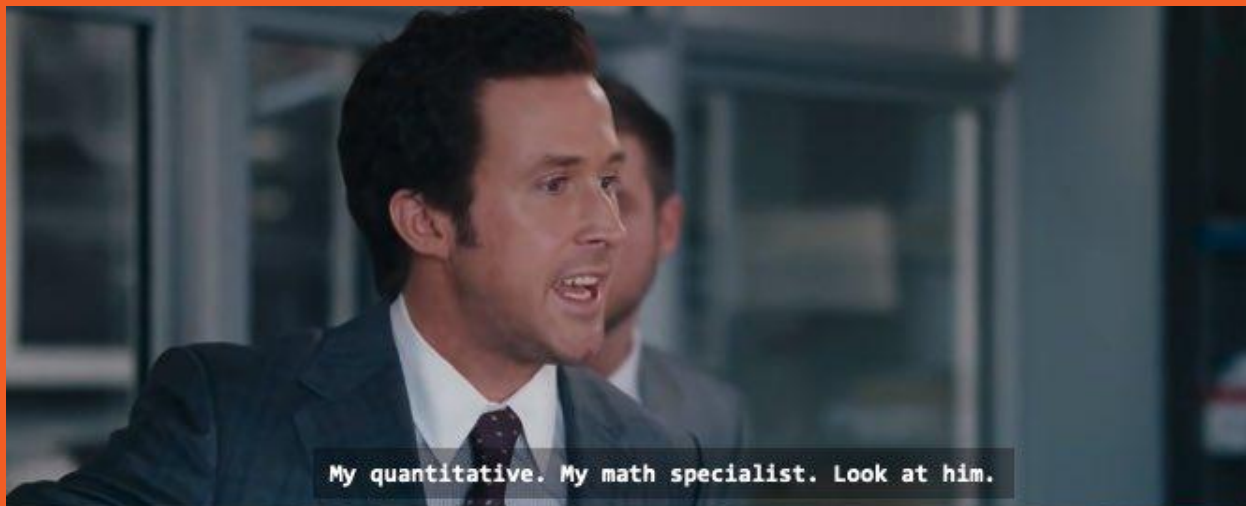




Quant Week 0

1/23/25





Agenda



1. Introductions
2. What is Quant?
3. Roles in Quant
4. Semester Overview
5. Onboarding



Introductions



General Introduction

- Name
- Year
- Major
- Where you're from
- Fun Fact

Brain Teaser - Bridge and Torch Problem



Bridge and Torch



- 4 people need to cross a river.
- Only 2 people can cross at a time
- Person A takes 10 minutes to cross, person B takes 5, C takes 2 and D takes 1.
- It's dark out so no one can cross without a torch.
- Only 1 torch.
- What is the fastest way to cross, and how long would it take?



Step 1: D & C Cross First

- D and C cross together, taking 2 minutes.
- Now C & D are on the other side.

Step 2: D Returns with the Torch

- D returns alone, taking 1 minute.
- Now A, B, and D are on the starting side, and C is on the other side.

Step 3: A & B Cross Together

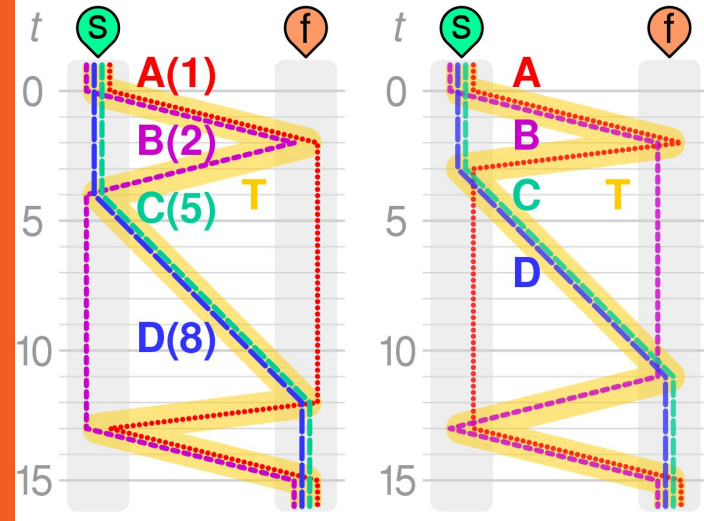
- A and B cross together, taking 10 minutes (since A is the slowest).
- Now A & B are on the other side, and D is still on the starting side.

Step 4: C Returns with the Torch

- C returns alone, taking 2 minutes.
- Now C & D are on the starting side again, and A & B are on the other side.

Step 5: C & D Cross Again

- C and D cross together, taking 2 minutes.
- Now everyone is on the other side.



What is “Quant”?



Quantitative Finance



- Quantitative Finance is essentially just the practice of using programmed models to predict asset prices
- The idea is simple to state, but is an incredible complex problem
- Within quantitative finance there are a variety of interesting areas and problems to solve
 - Different asset types
 - Different roles
 - Tons of data
 - Engineering challenges, like trade execution and speed
 - and massive implications



Asset Classes



- Stocks
 - High volatility stocks like GME, AMC, TSLA
- Derivatives/Options
- Foreign Exchange (Forex)
- Crypto
- Sports Betting

Key: high volatility, lots of price movement



Industry

Market Makers:



Market Maker

['mār-kət 'mā-kər]

A firm or individual who actively quotes two-sided markets in a particular security, providing bids and offers (known as asks) along with the market size of each.

 Investopedia

Goldman Sachs, Jane Street Capital, SIG
<https://www.investopedia.com/terms/m/marketmaker.asp>

Prop Trading Firms / HFTs:



Proprietary Trading

[prə-'prī-ə-,ter-ē 'trā-dīŋ]

A financial firm or bank that invests for direct market gain rather than earning commissions and fees by trading on the behalf of clients.

 Investopedia

Citadel, Optiver, IMC Trading, DRW, Virtu Financial, Chicago Trading Company, Wolverine Trading
<https://www.investopedia.com/terms/p/proprietarytrading.asp>

Roles in Quant



Roles in Quantitative Finance



Quantitative Traders

- Execute trades
- Teams divided based on asset class and other factors
- Act as a manager to automated trading systems
 - Change parameters
 - Respond to the market
- Analyze and improve decision making process in post-trade analysis
- Involved mathematics background



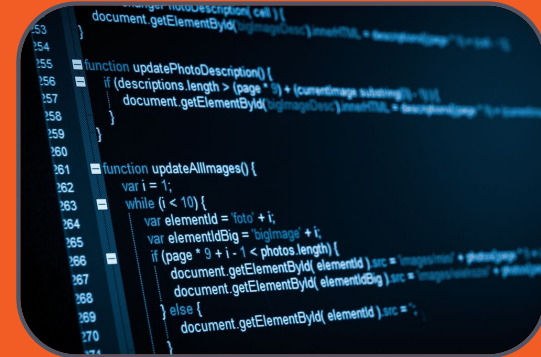
Much of young traders' time is spent shadowing other traders and developing what's referred to as "traders' intuition"

Roles in Quantitative Finance



Software Engineers

- Similar to their big tech counterparts
- Manage infrastructure
- Deal with scalability
- Optimize trading algorithms for speed and efficiency
 - Low latency programming
 - FPGAs



Generally working as a SWE in a quant firm is similar to working as a SWE anywhere else, but with higher stakes and an emphasis on efficient system design and speed.

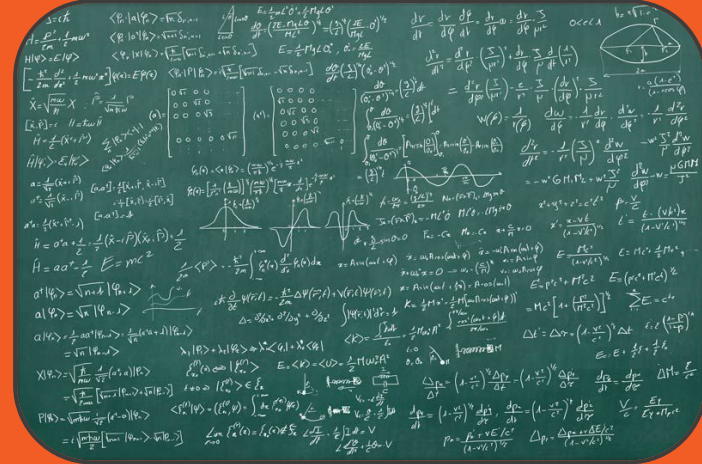
Roles in Quantitative Finance



Quantitative Researchers

- Work has aspects of both traders and engineers
- Analyze trades
- Use statistical inference and data analysis
- Develops algorithms at a higher level which will be optimized later
- Highly secretive and highly paid

Researchers generally have advanced degrees
(i.e. PhD and less frequently MS)



Semester Overview





Semester Overview



We plan on splitting the meetings into two parts

- Education
 - Trading Strategies
 - Market Making
 - High Frequency Trading
 - Interview Prep
 - Options Theory
- Project
 - This semester: Trading Algo Comp!

Note that this plan is dependent on general experience level and number of returning members.



Onboarding

Membership:

<https://tinyurl.com/y2fsu2u3>

Link to Onboarding Document,
Attendance, Slack and Team Folder

<https://tinyurl.com/2eptz2c4>



Project

Trading Strategy Competition!

Implement Trading algorithm(s) to attempt to beat the stock market

- Use finance package to pull stock market data
- Start with a base amount of money \$
- We won't tell you the time frame until the end - will give a week-long span
- Use any package you want! Pandas, NumPy, (PyTorch, Scikit-learn)

If you find any resources that require a subscription we can likely pay for it with club money! Do not hesitate

Rewards for 1st 2nd and 3rd Place (likely \$50 \$25 \$10)